

FCT SERVICE NOTE

SN 05090610a

Class A Service information enclosed.

Date: September 6, 2005

Subject: VCT M3 Detector service errata

Effectivity: All VCT systems

Details: Service Errata was not sent with M3 software release. Attached are the errata for Detector repair work.

Resolution:

Service procedure updates:

Many procedure updates and wording cleanup are in process. Check the Service Tech Pubs web page for the latest copy of the procedures.

General Tool issues:

You will need a 2.5 and 3mm hex drive bit for your torque driver. These will be supplied for systems with the M3A FMI and are being added to the Module FRU kits. Detector replacements need a 2.5mm bit for torquing the module screws.

The Deionizing fan is strongly suggested for module replacements but not mandatory.

Software replacement wizard utility:

The replacement wizard software is not fully functional in the M3 software release. It is fully functional in the M3A release. For the older M3 release, run the defined tools manually outside the wizard making sure to follow order of tests and calibrations as defined in the service instructions. Use the zalignmentWizard manually using the instructions given in the service doc. Run DasTools and calibrations manually.

Light seal plate screws: (Module replacements only)

There is a Torx T20 bit supplied with the module. This is used on the lightseal for most detectors. Early detectors used hex screws. Remove those screws and discard them. When installing the lightseal plate use the torx head screws supplied with the module kit. This change was made due to the fact that the hex screw heads can be deformed over use causing issues removing them. The torx head screws do not have this issue.

Light seal clips: (Module replacements only)

The light seal clips one either side of the module below the heat sink may or may not exist on your detector. Early detectors did not have these installed. Ignore steps dealing with these clips.

Module handle: (Module replacements only)

The module handle and other module tools will be shipping with systems and will be FMI'd to systems that do not already have them.

All tools except the handle is shipped with the modules. The handle is being added in the module kits separately at this time so you may or may not get one. The handle is not critical. Lift the module using the digital cable screws to remove it. Use the heat sink and/or module screws during new module insertion.

Bias tool notes: (Module replacements only)

If the bias tool does not attach well such as over top the heaters at either end of the detector, you may need to push on the bias tool near the magnet while torquing one of the module screws to make sure the alignment is maintained. You can see if the module is way out of alignment by looking along the front edge to see if the module looks in line with its neighbors. The bias tool has a redesign that fixes this issue for the future.

Calibrations:

Proper Calibrations must be performed in the following order.

- 1) Collimator Cal
- 2) Detailed Cal
- 3) Fastcal

There is some confusion over the need to run CT# adjust. CT# adjust is run automatically as part of detailed cal so it does not need to be run independently.

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